B.Tech. (CSE) 5th Semester (G-Scheme)
Examination, December–2025

FORMAL LANGUAGES & AUTOMATA

Paper–PCC-CSE-305-G

Time allowed : 3 hours] *[Maximum marks : 75*

Note : *Attempt five questions in all.* ***Question Number 1 is compulsory.*** *Select one question from each section.*

1. Explain the following: 2.5×6=15

   (i) Epsilon NFA

   (ii) FSM

   (iii) Regular Language

   (iv) GNF

   (v) Halting Problem

   (vi) Multitape Turing Machine

Section-A

2. Write down a short note on: 15

   (a) Minimization of finite automata

   (b) Acceptability of a string by a finite automata

3. Construct a Mealy Machine from the given transition table of Moore Machine. 15

Present State	Next State a = 0	Next State a = 1	λ
→ q_0	q_3	q_1	0
q_1	q_1	q_2	1
q_2	q_2	q_3	0
q_3	q_3	q_0	0

Section-B

4. State and prove Arden's method with example. 15

5. State and prove pumping lemma for regular languages. Also show that $L = \{a^i b^i c^i \{i \geq 1\}$ is not regular. 15

Section-C

6. Design a PDA for the language. 15

   $L = \{a^n b^n c^m \mid n \geq 1, m > 1\}$

7. Explain chomsky hierarchy of grammar in detail. 15

Section-D

8. What do you mean by Turing Machine? Explain the design and Halting problem of Turing Machine. 15

9. Write a short note on the following: 15

   (a) Undecidability

   (b) Rice's theorem